

Workshop on Deep Learning

Project Proposal

Group Number 13

- Elad Kulman - 206446361
- Tom Weitman - 326816600

End Product (System from User's Perspective)

The end product will include two separate features from a user perspective:

1. **Classic time series predictor** – every week (or different time interval - TBD) the model will predicate the performance of supported trading assets (BTC and TBD) for the next week (a number between -100 – 100)
2. **Feature classification** – the model will predict which feature is the most probable to be correlated with the asset price.

The system trains a baseline time-series model, but the core feature is what happens next. It runs tests to isolate and rank the news data, figuring out which specific narratives actually move the needle on the asset's price. Finally, the tool outputs clear visual charts that overlay the price predictions with the top-ranked news features, allowing the user to see exactly which headlines have the biggest impact on the market.

Training and Inference Schemes

The Architecture

We will utilize a lightweight Transformer-based time-series model. Rather than relying on a traditional RNN or LSTM, a Transformer is specifically chosen for its self-attention mechanisms. This allows us to mathematically extract and visualize exactly which inputs (e.g., specific news narratives vs. raw volume) the model prioritizes at any given time, fulfilling our core goal.

Training Scheme

The training pipeline is broken into two distinct phases:

1. **Baseline Supervised Training:** The model is trained on a merged dataset of daily price action and pre-processed news sentiment vectors to predict short-term price movements.
2. **Interpretability & Feature Ranking:** Once the baseline is established, we run an analysis pass using attention-weight extraction. We systematically mask out specific news features to calculate their exact impact on the model's accuracy, allowing us to rank the real-world narratives by their true market influence.

Inference Scheme

During execution, the inference pipeline takes in a rolling window of recent market data and news sentiment. However, the output is dual-purpose: the system outputs the standard price prediction, alongside a dynamic "importance breakdown" that flags the top three news features that directly drove that specific prediction.

Datasets

To ensure the project remains highly feasible and focused on model interpretability rather than data engineering, we will utilize static, pre-processed datasets.

- **Historical Market Data:** We will use standard OHLCV (Open, High, Low, Close, Volume, etc...) data for our target asset (Bitcoin) pulled from Yahoo Finance (yfinance) or other 3rd party source. This will be calculated into daily returns and basic volatility metrics to serve as our baseline numerical input.
- **News & Sentiment Data:** Rather than scraping raw articles, we will utilize an existing open-source sentiment dataset (such as the Kaggle Crypto News Sentiment Dataset or Financial PhraseBank). This provides us with thousands of historical financial headlines that are already labeled or vectorized, allowing us to focus on training and feature-ranking.

Compute and Storage Requirements

A major priority of this project's design is ensuring that our computational footprint fits entirely within the parameters of the school's Slurm GPU cluster and the provided Intel AI workstation. We will fine-tune an existing time-series model on approximately 10 years of past data (translating to thousands of data points), with each daily data point consisting of 10-100 features.

Third Party Tools and Models

- **Darts (Time Series Framework):** We will utilize the Darts Python library as our core machine learning framework. Darts provides a unified, highly efficient API

that streamlines data wrangling, model training, and backtesting. Crucially for our architecture, Darts natively supports the injection of external variables ("past covariates"), making it the ideal framework to align our LLM-generated sentiment vectors with historical price sequences.

- Temporal Fusion Transformer (TFT): For our predictive network, we will implement the TFT architecture (accessed via Darts). TFT is a state-of-the-art deep learning model explicitly designed for multimodal time-series forecasting.

Existing Solutions and Related Work

Our project builds upon several existing foundations in financial machine learning, but addresses a critical gap in model transparency:

- Financial NLP Models: Existing open-source models like FinBERT, FinGPT have successfully proven that pre-trained language models can accurately classify financial sentiment (positive, negative, neutral) from headlines and earnings reports. However, these models typically stop at text classification and do not integrate directly into time-series forecasting.
- Time-Series Transformers: Architectures like Informer or Autoformer have adapted the Transformer attention mechanism specifically for complex time-series forecasting. While highly accurate, they are often treated as black boxes in algorithmic trading, leaving researchers blind to the exact external variables driving the predictions.
- Quantitative Investment Platforms (Qlib): Frameworks such as Microsoft's Qlib provide comprehensive pipelines for quantitative research, excelling at data ingestion and baseline model training (e.g., LightGBM) across massive equity panels.
- Financial Reinforcement Learning (FinRL): Libraries like FinRL are designed to train autonomous agents that execute specific market actions (Buy, Sell, Hold) to maximize reward. While powerful for building automated trading bots, this reinforcement learning paradigm falls outside our project's scope.